ePoster ID: R363

Sarbjit Saini | ssaini@jhmi.edu

REMIX-1/-2: Early Symptom Improvements With Remibrutinib in Chronic Spontaneous Urticaria From Week 1

Sarbjit Saini,[1] Robert Szalewski,[2] Xinghua Gao,[3] Sabine Altrichter,[4–7] Sibylle Haemmerle,[8] Noriko Seko,[9] Michihiro Hide[10,11]

[1]Johns Hopkins Asthma and Allergy Center, Baltimore, MD, USA; [2]Allergy, Asthma, and Immunology Associates PC, Lincoln, NE, USA; [3]The First Hospital of China Medical University, Shenyang, China; [4]Department of Dermatology and Venerology, Kepler University Hospital, Linz, Austria; [5]Center for Medical Research, Johannes Kepler University, Linz, Austria; [6]Institute for Allergology, Charité-Universitätsmedizin Berlin, Berlin, Germany; [7]Fraunhofer Institute for Translational Medicine and Pharmacology ITMP, Berlin, Germany; [8]Novartis Pharma AG, Basel, Switzerland; [9]Novartis Pharma KK, Tokyo, Japan; [10]Hiroshima City Hiroshima Citizens Hospital, Hiroshima, Japan; [11]Hiroshima University, Hiroshima, Japan

KEY FINDINGS & CONCLUSIONS

- In REMIX-1 and REMIX-2, fast and significant improvements in CSU symptoms of itch and hives vs placebo were observed as early as week 1, with further improvements through week 24
- Likewise, ≥50% of patients with CSU achieved MID for UAS7, ISS7, and HSS7, by weeks 1 to 2 of treatment with remibrutinib
- A limitation of the study is that post hoc analyses were used which may introduce biases from selective data analysis
- Overall, remibrutinib has the potential to become a novel oral treatment option with an early onset of response in patients with CSU inadequately controlled by H_1-AH

Scan to obtain:
- Poster

https://www.medicalcongressposters.com//Default.aspx?doc=664fe

Copies of this poster obtained through the Quick Response (QR) code are for personal use only and may not be reproduced without permission of the authors.

This study was sponsored by Novartis Pharma AG, Basel, Switzerland.

Poster presented at the American College of Allergy, Asthma & Immunology (ACAAI) Annual Scientific Meeting, held in Boston, MA, USA, on October 24–28, 2024.

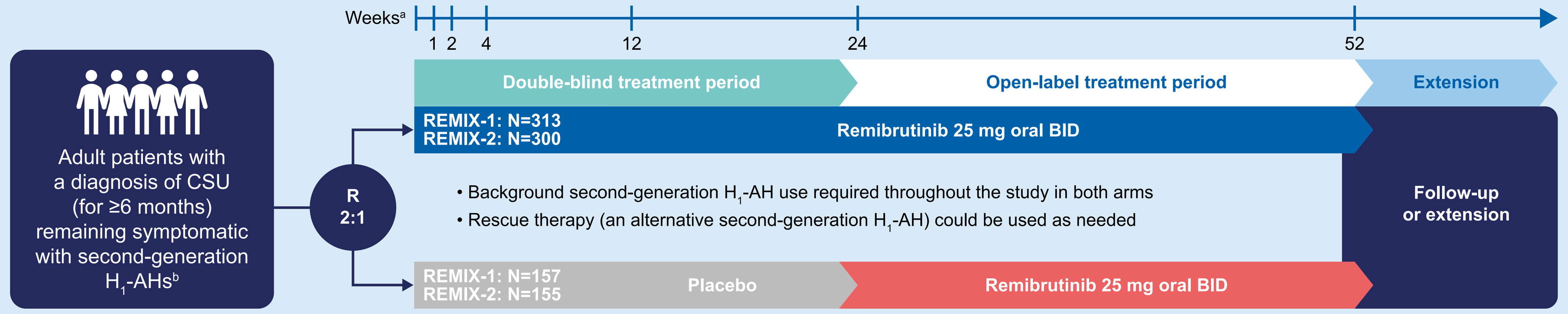
INTRODUCTION

- CSU is a disease which is characterized by the spontaneous appearance of itchy wheals, angioedema, or both, lasting ≥6 weeks and can substantially impact patients' well-being[1–3]
- First-line treatment involves second-generation H_1-AHs; however over 50% of patients do not achieve symptom control with H_1-AHs alone[3,5,6]
- Remibrutinib is a novel, oral, highly selective BTK inhibitor that prevents mast cell–mediated release of histamine and other proinflammatory mediators[4,7,8]
  - Remibrutinib has shown superior efficacy vs placebo in symptom improvement after 12 weeks of treatment in patients with CSU[4]
- A UAS7 decrease of 9.5–10.5 points is commonly considered to indicate the MID[9]
- The objective of this analysis was to assess early treatment responses with remibrutinib vs placebo in the phase 3 REMIX-1 (NCT05030311) and REMIX-2 (NCT05032157) studies

METHODS

- REMIX-1 and REMIX-2 are multicenter, randomized, double-blind, parallel-arm, placebo-controlled phase 3 studies investigating the safety and efficacy of remibrutinib administered orally
- Patients were randomized 2:1 to remibrutinib 25 mg twice daily, or to placebo, for 24 weeks (**Figure 1**)
- In this analysis, LS mean CFB in weekly UAS7, ISS7, and HSS7 was assessed at weeks 1, 2, 4, 12, and 24 (weeks 1, 2, and 4 were post hoc analyses)
  - A higher score reflects higher disease activity
- The onset of action of remibrutinib was explored post hoc in terms of early achievement of MID in CSU disease activity, defined as a change in score of ≥10.5 for UAS7, ≥5.0 for ISS7, and ≥5.5 for HSS7[9]
- Data were analyzed using summary statistics, and MMRM was used for the analysis of CFB

Figure 1. Study Design for REMIX-1 and REMIX-2 Studies



[a]UAS7, ISS7, and HSS7 scores were collected weekly and analyzed at weeks 1, 2, 4, 12, and 24.
[b]Presence of itch and hives for ≥6 consecutive weeks prior to screening despite the use of a second-generation H_1-AH; UAS7 score ≥16, ISS7 score ≥6, and HSS7 score ≥6 during the 7 days prior to randomization (day 1).
BID, twice daily; CSU, chronic spontaneous urticaria; H_1-AH, histamine 1-antihistamine; HSS7, weekly hives severity score; ISS7, weekly itch severity score; R, randomized; UAS7, weekly urticaria activity score.

RESULTS

- Overall, 470 patients were randomized in REMIX-1 (remibrutinib, n=313; placebo, n=157) and 455 patients in REMIX-2 (remibrutinib, n=300; placebo, n=155) (**Table 1**)

Demographics and Clinical Characteristics

- Patient demographics and baseline characteristics were well balanced between treatment arms in REMIX-1 and REMIX-2 (**Table 1**)
  - Baseline weekly UAS7, ISS7, and HSS7 were similar between remibrutinib and placebo in REMIX-1 and REMIX-2

This table has been published in the New England Journal of Medicine (Metz M, et al. *N Engl J Med.* 2025;392(10):984-94). For questions or requests regarding its use, please contact the Global Medical team.

Change From Baseline in UAS7, ISS7, and HSS7, Week 1 to Week 24

- The full analysis set included 462 and 450 patients randomized in REMIX-1 and REMIX-2, respectively
- Remibrutinib showed significant improvements vs placebo in LS mean (±SE) CFB-UAS7 as early as week 1 (**Figure 2**)
  - REMIX-1: −11.3 ± 0.6 vs −4.0 ± 0.8 (P<0.001)
  - REMIX-2: −11.3 ± 0.5 vs −2.9 ± 0.7 (P<0.001)
- Significant improvements were also shown in LS mean (±SE) CFB-ISS7 and CFB-HSS7 from week 1 (**Figure 2**)
  - CFB-ISS7:
    - REMIX-1: −5.2 ± 0.3 vs −2.1 ± 0.4 (P<0.001)
    - REMIX-2: −5.0 ± 0.3 vs −1.4 ± 0.3 (P<0.001)
  - CFB-HSS7:
    - REMIX-1: −6.0 ± 0.3 vs −1.9 ± 0.4 (P<0.001)
    - REMIX-2: −6.3 ± 0.3 vs −1.5 ± 0.4 (P<0.001)
- Improvements in LS mean (±SE) UAS7, ISS7, and HSS7 were further improved and maintained to week 24 with remibrutinib (**Figure 2**)

MID of UAS7, ISS7, and HSS7 at Week 1

- The number of patients who achieved MID in disease activity for UAS7, ISS7, and HSS7 is displayed in **Figure 3**
- A higher proportion of patients on remibrutinib compared with placebo achieved improvements in UAS7, ISS7, and HSS7 by week 1 (50.7% vs 14.5%; 50.4% vs 17.2%; 52.1% vs 17.2%), indicating early onset of action of remibrutinib

This figure has been published in the New England Journal of Medicine (Metz M, et al. *N Engl J Med.* 2025;392(10):984-94). For questions or requests regarding its use, please contact the Global Medical team.

This figure has been published in the New England Journal of Medicine (Metz M, et al. *N Engl J Med.* 2025;392(10):984-94). For questions or requests regarding its use, please contact the Global Medical team.

Acknowledgments

Medical writing support and editorial support were provided by Rhianna Hill, PhD (BOLDSCIENCE Ltd. UK), and were funded by Novartis Pharmaceuticals Corporation. This presentation was developed in accordance with Good Publication Practice (GPP) guidelines. The authors had full control of the content and made the final decision on all aspects of this publication.

Abbreviations

BID, twice daily; BMI, body mass index; BTK, Bruton's tyrosine kinase; CFB, change from baseline; CI, confidence interval; CSU, chronic spontaneous urticaria; H_1-AH, histamine 1-antihistamine; HSS7, weekly hives severity score; ISS7, weekly itch severity score; LS, least squares; MID, minimum important difference; MMRM, mixed models for repeated measures; R, randomized; SD, standard deviation; SE, standard error; UAS7, weekly urticaria activity score.

Disclosures

SS has received grant/research/clinical trial support from the National Institutes of Health, Novartis, Allakos, Incyte, and Jasper and is a consultant/advisory board member for Allakos, Granular Therapeutics, Novartis, Aquestive, GSK, Celldex, Evommune, Regeneron, Escient, Innate, Celltrion, and Sanofi. **RS** serves as a speaker for AstraZeneca and Pharming and has received clinical trial support and/or consulting fees from Genentech, Sanofi, Arcutis, AstraZeneca, Areteia, GlaxoSmithKline, AbbVie, Amgen, Novartis, and Regeneron. **XG** has received consultancy/speaker honoraria from Novartis, Pfizer, Astellas, Galderma, Janssen, Sanofi, GSK, LEO, Lilly, MEDA Pharma (a Mylan Company), and has acted as scientific advisory board member for Novartis, Lilly, Sanofi, MEDA Pharma (a Mylan Company). **SA** is a speaker and/or advisor for and/or has received research funding from Allakos, AstraZeneca, BioCryst, Blueprint, CSL Behring, Incyte, Moxie, Novartis, Pharvaris, Sanofi, Takeda, and Thermo Fisher. **SH** is an employee of Novartis Pharma AG, Basel, Switzerland. **NS** is an employee of Novartis Pharma KK, Tokyo, Japan. **MH** has received lecture and/or consultation fees from Kaken Pharmaceutical, Kyowa Kirin, Mitsubishi Tanabe Pharma, MSD, Novartis, Sanofi, TAIHO Pharmaceutical, and Teikoku Seiyaku.

References

1. Zuberbier T, et al. *Allergy*. 2022;77:734–766.
2. Gonçalo M, et al. *Br J Dermatol*. 2021;184:226–236.
3. Maurer M, et al. *Allergy*. 2017;72:2005–2016.
4. Maurer M, et al. *J Allergy Clin Immunol*. 2022;150:1498–1506.e2.
5. Yosipovitch G, et al. *Dermatol Ther (Heidelb)*. 2023;13:1647–1660.
6. Bernstein J, et al. Presented at ACAAI, November 9–13, 2023; Anaheim, CA. Abstract D006.
7. Angst D, et al. *J Med Chem*. 2020;63:5102–5118.
8. Kaul M, et al. *Clin Transl Sci*. 2021;14:1756–1768.
9. Mathias SD, et al. *Allergy Asthma Proc*. 2015;36:394–398.